The minimum number of deferred-acceptance states in a doubly prime instance

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Abstract

For every $n \geq 2$, the minimum number of reachable serial deferred-acceptance states among doubly prime strict complete stable-marriage instances of order n is $3 \cdot 2^{n-1} - 1$. The lower bound follows by selecting one rejection witness for each nonfirst proposal. The resulting connected directed acyclic graph has two incoming neighbors at each nonsource, and every one of its order ideals is a reachable state. A counting lemma for such graphs supplies the bound. A rejection-chain family attains it with $2n - 1$ proposals. This proves the conjecture in Ishida's Section 7(A).

1 The question and its conventions

An instance has n proposers P, n receivers R, and strict complete preference lists. A serial deferred-acceptance state is (h, q), where $q(p)$ counts proposals made by p and $h(r)$ is the proposer held by r, or $\perp$. Initially all counts are zero and all receivers are empty. A legal step selects an unheld proposer with an untried receiver and makes its next proposal. The receiver keeps its preferred of the new proposer and its previous holder. Write $\mathrm{Reach}(I)$ for all states reached by all finite legal schedules, including the initial and terminal states.

An *input-independent partition* pairs partitions $P = P_1 \sqcup \cdots \sqcup P_t$ and $R = R_1 \sqcup \cdots \sqcup R_t$ with $|P_j| = |R_j| > 0$ for every j, such that every member of P_j ranks all of R_j above every receiver outside R_j, and every member of R_j ranks all of P_j above every proposer outside P_j. An instance is *input-prime* if it has no such partition with $t \geq 2$. Number a proposer's events from 1, and put

$$F(q) = \{(p, k) : 1 \leq k \leq q(p)\}, \qquad \mathcal{F}(I) = \{F(q) : (h, q) \in \mathrm{Reach}(I)\}, \qquad E(I) = \bigcup \mathcal{F}(I).$$

The change from zero-based event indices in [1] is only a relabeling. The instance is *behaviorally prime* if no partition $E(I) = E_1 \sqcup E_2$ into nonempty sets satisfies

$$\mathcal{F}(I) = \{X_1 \cup X_2 : X_j \in \mathcal{F}(I)|_{E_j},\ j = 1, 2\}, \qquad \mathcal{F}(I)|_{E_j} = \{X \cap E_j : X \in \mathcal{F}(I)\}.$$

It is *doubly prime* if both properties hold. Let

$$m_n = \min\{|\,\mathrm{Reach}(I)| : I \text{ is doubly prime of order } n\}, \qquad \min \varnothing = +\infty.$$

Theorem 1. *For every $n \geq 2$, $m_n = 3 \cdot 2^{n-1} - 1$. A minimizing instance can simultaneously have total proposal count $2n - 1$.*

Ishida [1, Proposition 21.4 and Section 7(A)] proves the lower bound $2^n + n - 1$ and asks for the equality above. The rejection-chain construction in Section 4 already appears in the

research attempts for catalogue entry C78-352;[1] we include its proof to make attainability explicit. The contribution here is the universal matching lower bound. In particular, we do not use the experimentally checked converse relating proposal overlap to behavioral factors discussed in [1, Sections 2 and 4].

2 Counting ideals by component merges

For a finite directed acyclic graph D, an *ideal* is a vertex set closed under predecessors. Let $\mathcal{J}(D)$ be its set of ideals. Connectivity below refers to the underlying undirected graph.

Lemma 2. *If D is connected, has $n \geq 1$ sources, and every nonsource has at most two incoming neighbors, then*

$$|\mathcal{J}(D)| \geq 3 \cdot 2^{n-1} - 1.$$

Proof. Choose a topological order placing all sources first. After these n vertices have been inserted, the processed graph has n singleton components. Every later vertex attaches to one or two existing components. Exactly $n-1$ vertices merge two distinct components, since the final graph is connected. Every processed component contains a source.

For a merging vertex v, let $s(v)$ be the number of sources in the two components being merged. Every ancestor of v lies in those components. Let $A(v)$ consist of v and all its ancestors. For every subset T of the $n - s(v)$ sources outside those components, $A(v) \cup T$ is an ideal. These $2^{n-s(v)}$ ideals are distinct, since their intersection with the outside sources recovers T. Their latest nonsource in the fixed topological order is v. Thus the families for different merging vertices are disjoint, and none consists solely of sources. All 2^n subsets of the sources are themselves ideals.

Record each component merger as a binary node, with the original source components as leaves. Vertices that do not merge components leave this record unchanged. The result is a full binary tree T with n leaves; $s(v)$ is the number of descendant leaves of its internal node v. For a full binary tree T with ℓ leaves, define

$$C(T) = \sum_{v \text{ internal}} 2^{\ell - s(v)}.$$

For the merger tree $\ell = n$, so $C(T)$ is the total size of the families above. We claim $C(T) \geq 2^{\ell-1} - 1$. For a single leaf both sides are zero. If the root has subtrees T_a and T_b with $a, b \geq 1$ leaves, where $a + b = \ell$, then the root contributes 1, each internal node v of T_a contributes $2^{\ell - s(v)} = 2^b \cdot 2^{a - s(v)}$, and symmetrically for T_b. Induction gives

$$\begin{aligned} C(T) &= 2^b C(T_a) + 2^a C(T_b) + 1 \\ &\geq 2^\ell - 2^a - 2^b + 1 \geq 2^{\ell-1} - 1. \end{aligned}$$

The last inequality follows from $(2^a - 2)(2^b - 2) \geq 0$, equivalently $2^a + 2^b \leq 2^{\ell-1} + 2$. Adding the source-only ideals proves the lemma. □

3 Rejection witnesses give reachable ideals

We first record the execution facts needed to apply Lemma 2. A receiver always holds its favorite among its past proposers. Thus q determines h, and distinct event sets determine distinct states. Every run has at most n^2 steps. Two enabled steps by different proposers

[1] See the GPT 6 Astra attempt, "A chain of rejections", and the Opus 5.5 attempt, "The upper bound for every n". Both leave the general lower bound unresolved.

commute and leave each other enabled: another proposer's action cannot make an unheld proposer held. If they contact the same receiver, its holder after both steps is the best of the old holder and the two newcomers.

Consequently the terminal state is unique. For completeness, induct on $n^2 - \sum_p q(p)$: two distinct first steps have a common successor after commuting; the inductive uniqueness of terminal descendants then identifies the terminal states of both branches. Every prefix extends to this terminal state. Denote its counts by q^* and its events by E. Every proposer is held there. Otherwise an unheld proposer must have exhausted its list, so all n receivers have received proposals and hold n distinct proposers, contradicting the existence of an unheld one. In particular, all n first events occur.

Let H be the graph on proposers joining those who contact a common receiver among the events in E. If H were disconnected, events in different components would address disjoint sets of receivers. Deleting the other components' steps from a legal prefix preserves legality; conversely, any projected prefixes can be interleaved because their receiver updates are disjoint. Every prefix uses only E, since it extends to the unique terminal state. Hence the full event family is exactly the product of its component projections, each nonempty. Behavioral primality therefore implies that H is connected.

Lemma 3. *For a behaviorally prime order-n instance, there is a connected directed acyclic graph D on E with exactly n sources and exactly two incoming neighbors at each nonsource, such that every ideal of D is a distinct reachable state.*

Proof. Fix a complete legal execution and write $e(p, k)$ for its kth proposal by p. For each $k > 1$, let r be the receiver contacted by $e(p, k-1)$. Immediately before $e(p, k)$, proposer p is unheld and r holds some w it prefers to p. Choose w's earlier proposal $e(w, j)$ to r as a witness, and add the edges

$$e(p, k-1) \longrightarrow e(p, k), \qquad e(w, j) \longrightarrow e(p, k).$$

Both edges point forward in the fixed execution. Their tails are distinct because $w \neq p$. The sources are exactly the first events.

Let X be an ideal and execute its events in topological order. Inductively suppose the preceding events form a legal prefix. The own-proposer edges ensure that, when $e(p, k)$ is next, p has made exactly its first $k-1$ proposals. For $k = 1$ it is unheld. For $k > 1$, the witness has already proposed to p's previous receiver, whose holder only improves; that receiver therefore cannot hold p. Neither can an earlier receiver: p was unheld before its following proposal, and another proposer's action cannot make it held again. Thus p is unheld and its next proposal is legal. This proves reachability of X. Distinct ideals give distinct proposal counts and hence distinct states.

It remains to prove connectivity. Contract each proposer's own event chain. Fix a receiver r. Every proposer p who contacted r except its final holder has a subsequent own proposal: otherwise p would be unheld at termination. Its chosen witness at r is some proposer w strictly preferred to p, and the contracted graph contains pw. Iterating such strictly improving witness links among the finite set of proposers to r reaches its best proposer, the final holder. Therefore all proposers who contacted r are connected in the contraction. This connects the endpoints of every edge of H. Since H is connected, so are the contraction and D. □

The witness graph need not describe every legal prefix: choosing one witness selects sufficient prerequisites. Its ideals form a subfamily of the reachable event family, which is all the lower bound requires. Lemmas 2 and 3 prove

$$|\operatorname{Reach}(I)| \geq 3 \cdot 2^{n-1} - 1$$

for every behaviorally prime instance, and in particular every doubly prime instance. The same merger argument gives $|E| \geq 2n-1$: besides the n sources there are $n-1$ distinct merging vertices.

Corollary 4. *Let I be a behaviorally prime instance of order n, and let $Q^* = \sum_p q^*(p) = |E|$ be its total number of proposals. Then $Q^* \geq 2n-1$ and*

$$|\operatorname{Reach}(I)| \geq 3 \cdot 2^{n-1} - 1 + (Q^* - (2n-1)).$$

In particular, $|\operatorname{Reach}(I)| = 3 \cdot 2^{n-1} - 1$ implies $Q^* = 2n-1$.

Proof. Let D be the graph of Lemma 3; its vertex set is E, and it satisfies the hypotheses of Lemma 2. Fix a topological order and the merging vertices as in the proof of Lemma 2. As noted above, the n sources and the $n-1$ merging vertices are distinct, so $Q^* \geq 2n-1$; for doubly prime instances this is also [1, Proposition 21.4]. Each of the remaining $Q^* - (2n-1)$ nonsources u merges no components. The set $A(u)$ consisting of u and its ancestors is an ideal, and its latest nonsource in the fixed order is u, since every ancestor of u precedes u. These ideals are therefore distinct. None of them occurs among the ideals counted in the proof of Lemma 2, each of which either consists of sources or has a merging vertex as its latest nonsource. Hence $|\mathcal{J}(D)| \geq 3 \cdot 2^{n-1} - 1 + (Q^* - (2n-1))$, and Lemma 3 gives the same bound for $|\operatorname{Reach}(I)|$. If $|\operatorname{Reach}(I)| = 3 \cdot 2^{n-1} - 1$, this bound gives $Q^* \leq 2n-1$, and equality follows. □

4 Attainment and double primality

For $n \geq 2$, take proposers $p_0, p_1, \ldots, p_{n-1}$ and receivers $r_1, \ldots, r_n$. Let p_0 rank $r_1, \ldots, r_n$ in that order. For $1 \leq i < n$, let p_i and r_i rank each other first. Complete every unspecified list arbitrarily to a strict complete order.

Each p_i with $i > 0$ makes one proposal and is held permanently at r_i. Only p_0 makes later proposals. At $q(p_0) = k-1$, its kth proposal, for $k \geq 2$, requires rejection from r_{k-1} by p_{k-1}. It follows that the reachable pointers are exactly

$$\begin{array}{ll} q(p_0) = 0: & q(p_i) \in \{0,1\} \quad (i > 0); \\ q(p_0) = k,\ 1 \leq k \leq n: & q(p_i) = 1 \quad (1 \leq i < k), \qquad q(p_i) \in \{0,1\} \quad (k \leq i < n). \end{array}$$

For sufficiency, execute the selected first proposals of the p_i, then the first k proposals of p_0, stopping immediately after the last one, even if rejected. Consequently

$$|\operatorname{Reach}(I_n)| = 2^{n-1} + \sum_{k=1}^{n} 2^{n-k} = 3 \cdot 2^{n-1} - 1.$$

To check behavioral primality directly, name p_0's events $a_1, \ldots, a_n$ and p_i's event b_i. The displayed pointers say that the event family is precisely the ideals of

$$a_1 < a_2 < \cdots < a_n, \qquad b_i < a_{i+1} \quad (1 \leq i < n).$$

A mandatory predecessor e of an event f cannot lie in a different factor of an event-family product: combine a projection containing f with the empty projection in e's factor to contradict that requirement. All events occur, and the displayed predecessor edges connect them all. Thus no nontrivial product exists.

For input primality, suppose the block containing p_0 has size $k < n$. Its receivers must be $r_1, \ldots, r_k$, the first k choices of p_0. Since r_i ranks p_i first, that block must contain each $p_1, \ldots, p_k$ as well as p_0, contradicting its size k. Therefore I_n is doubly prime. This proves

attainability and completes Theorem 1. Its terminal counts are $q(p_0) = n$ and $q(p_i) = 1$ for $i > 0$, giving $2n - 1$ proposals. The same instance thus attains the proposal lower bound of [1, Proposition 21.4] and the state minimum.

Ishida [1, Section 7(A)] observes that every minimizer found also attains $Q^* = 2n - 1$ and asks whether the two minima are always achieved together. The source does not say whether only this direction is meant or also its converse, so we answer both. Every doubly prime instance with m_n states has exactly $2n - 1$ proposals, since by Corollary 4 an instance with $Q^* \geq 2n$ has at least $3 \cdot 2^{n-1}$ states. The converse, that every doubly prime instance with $2n - 1$ proposals has m_n states, fails for $n = 3$. Let p_1 and p_2 rank r_1, r_2, r_3, let p_3 rank r_1, r_3, r_2, and let every receiver rank p_1 first, p_2 second and p_3 third. Then p_1 is held at r_1 from its first proposal on. Proposers p_2 and p_3 propose first to r_1, then to r_2 and r_3 respectively. A third proposal by p_2 (by p_3) needs an earlier rejection at r_2 (at r_3), which only an earlier third proposal by p_3 (by p_2) can cause, since p_1 never addresses these receivers; hence no third proposal occurs. Write a for the proposal of p_1, b_1, b_2 for those of p_2, and c_1, c_2 for those of p_3. Since r_1 holds its favorite past proposer, b_2 can be made exactly when a and b_1 have been made, and c_2 exactly when c_1 and at least one of a and b_1 have been made. Hence the reachable event sets are exactly the subsets of $\{a, b_1, b_2, c_1, c_2\}$ obeying these two conditions; for sufficiency, make the selected first proposals before the second ones. There are $2 + 3 + 3 + 6 = 14$ of them, according as they contain neither of a, b_1, only a, only b_1, or both. The terminal event set is $\{a, b_1, b_2, c_1, c_2\}$, so $Q^* = 5$. Every receiver ranks p_1 first, so every block P_j of an input-independent partition contains p_1; hence $t = 1$ and the instance is input-prime. Since a and b_1 are mandatory predecessors of b_2, and c_1 of c_2, the argument above shows that an event-family product could only have the factors $\{a, b_1, b_2\}$ and $\{c_1, c_2\}$. It would then contain $\{c_1, c_2\}$, the union of the projections of $\varnothing$ and $\{a, c_1, c_2\}$, which is not reachable. So the instance is doubly prime, with $2n - 1 = 5$ proposals and $14 > m_3 = 11$ states.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] Y. Ishida. *Prime factorisation of stable-matching instances: uniqueness, simultaneous products, and an exact census.* arXiv:2610.05617v1, 2026. https://arxiv.org/html/2610.05617v1.